

Sisyphus Cooling — Minimum Review Note

friction, kinetic theory, diffusion, and temperature limit

Physical mechanism. Two ground substates experience shifted periodic potentials $V^\pm(z)$. An atom climbs a hill, is optically pumped near the top into the other substate, and loses potential energy. Repetition produces cooling. The key is a velocity-dependent lag of the internal populations behind their local steady-state values.

Symbols and regime.

$$q = \frac{2\pi}{\lambda}, \quad \mathbf{p} = m\mathbf{v}, \quad \tau_p^{-1} = \Gamma_{+-} + \Gamma_{-+}, \quad E_r = \frac{\hbar^2 q^2}{2m}.$$

Main approximations:

$$v\tau_p \ll \lambda, \quad k_B T \gg |V^\pm|,$$

plus stationary, semiclassical motion and order-of-magnitude accuracy.

1. Quick friction estimate. In one pumping time the atom moves $\Delta z = v\tau_p$. Both the net pumping bias and the energy climbed are reduced by $\Delta z/\lambda$:

$$\Delta\Gamma \sim \tau_p^{-1} \frac{v\tau_p}{\lambda}, \quad \Delta E \sim V_0 \frac{v\tau_p}{\lambda},$$

$$\frac{dE}{dt} \sim -\Delta\Gamma \Delta E \sim -\frac{V_0 v^2 \tau_p}{\lambda^2} = -\frac{E}{\tau_{\text{cool}}},$$

$$\tau_{\text{cool}}^{-1} \sim \frac{V_0 \tau_p}{m\lambda^2} \sim -\frac{\delta}{\Gamma_e} \frac{E_r}{\hbar}.$$

For red detuning $\delta < 0$, the cooling rate is positive. Since $V_0 \propto I$ and $\tau_p \propto I^{-1}$, the rate is approximately independent of laser intensity (unlike Doppler cooling).

2. Transport equation. For $\epsilon_p^\pm = p^2/(2m) + V^\pm(\mathbf{r})$,

$$\frac{\partial f_p^\pm}{\partial t} + \underbrace{\mathbf{v} \cdot \nabla_r f_p^\pm}_{\text{streaming}} + \underbrace{\mathbf{F}^\pm \cdot \nabla_p f_p^\pm}_{\text{acceleration}} = \left(\frac{\partial f_p^\pm}{\partial t} \right)_{\text{pump}},$$

$$\mathbf{F}^\pm = -\nabla V^\pm.$$

Hamiltonian flow has zero phase-space divergence (Liouville's theorem); the right side is a source for each substate but conserves $f^+ + f^-$.

For a thermal momentum distribution,

$$\nabla_r f \sim \frac{f}{\lambda}, \quad |\mathbf{F}| \sim \frac{|V|}{\lambda}, \quad \nabla_p f = -\frac{\mathbf{v}}{k_B T} f,$$

so

$$\frac{|\mathbf{F} \cdot \nabla_p f|}{|\mathbf{v} \cdot \nabla_r f|} \sim \frac{|V|}{k_B T} \ll 1.$$

In a stationary state, neglecting the force-induced correction to f gives

$$\mathbf{v} \cdot \nabla_r f_p^\pm = \left(\frac{\partial f_p^\pm}{\partial t} \right)_{\text{pump}}.$$

3. Pumping and local steady state.

$$\left(\frac{\partial f_p^+}{\partial t} \right)_{\text{pump}} = -\Gamma_{-+} f_p^+ + \Gamma_{+-} f_p^-,$$

$$\frac{\bar{f}_p^+}{\bar{f}_p^-} = \frac{\Gamma_{+-}}{\Gamma_{-+}},$$

$$\frac{\bar{f}_p^+ - \bar{f}_p^-}{\bar{f}_p^+ + \bar{f}_p^-} = -\sin(2qz).$$

Let $f_p^+ + f_p^- = 2f_0$ and write $f_p^\pm = \bar{f}_p^\pm \pm \delta f$. Motion through the inhomogeneous field creates the lag

$$\delta f = 2qv_z \tau_p f_0 \cos(2qz).$$

It is odd in v_z : atoms moving in opposite directions have opposite population corrections.

4. Lag $\rightarrow$ friction. Weighting $-\nabla V^\pm$ by the two populations and averaging over one period, the local-steady-state part vanishes; the lag remains:

$$\langle F_z \rangle = -2q^2 V_0 \tau_p v_z = -\alpha v_z, \quad \alpha = 2q^2 V_0 \tau_p.$$

Thus $\langle F_z v_z \rangle < 0$. More generally,

$$\frac{1}{\tau_{\text{cool}}} = \frac{4q^2 V_0}{m} \frac{\int d^3p f_0 v_z^2 \tau_p}{\int d^3p f_0 v_z^2},$$

in agreement with the quick estimate up to numerical factors.

5. Heating and steady temperature. The periodic force also produces random momentum changes:

$$|F| \sim \frac{4\pi V_0}{\lambda}, \quad \Delta p \sim F\tau_p,$$

$$D_p \sim \frac{(\Delta p)^2}{\tau_p} \sim \frac{(4\pi)^2 V_0^2 \tau_p}{\lambda^2}.$$

At steady state, diffusion balances friction:

$$\langle p_z^2 \rangle \sim D_p \tau_{\text{cool}}, \quad E_{\text{steady}} \sim \frac{D_p \tau_{\text{cool}}}{m} \sim V_0.$$

Hence lowering intensity lowers the semiclassical temperature even though the cooling rate is nearly intensity independent.

6. Quantum limit. Localization to an optical well gives $\Delta p \sim \hbar/\lambda$ and an energy scale of order E_r . When $V_0 \lesssim E_r$, the classical treatment fails; reducing intensity further does not yield arbitrarily low energy. Detailed calculations give roughly

$$E_{\text{min}} \sim 30E_r \text{ (1D)}, \quad E_{\text{min}} \sim 40E_r \text{ (3D)},$$

$$V_0^{\text{opt}} \sim 50E_r.$$

Sub-recoil methods instead accumulate atoms in weakly coupled, narrow-velocity (“dark”) states.

One-line chain.

$$\begin{aligned} \text{motion} &\rightarrow \text{population lag} \rightarrow F = -\alpha v \rightarrow \text{cooling}, \\ \text{diffusion} &\rightarrow E \sim V_0, \quad \text{quantum motion} \rightarrow E_r. \end{aligned}$$

Self-check (cover the key below).

1. Why do two factors $v\tau_p/\lambda$ enter the quick estimate?
2. What assumptions reduce Eq. (4.78) to Eq. (4.84)?
3. Why does the exact local steady state give zero spatially averaged force?
4. What makes the surviving force proportional to $-v_z$?
5. Why can the cooling rate be intensity independent while E_{steady} is not?
6. What invalidates the classical prediction as V_0 becomes small?

Key. (1) Small pumping bias and small fraction of the hill climbed. (2) Stationarity and $|V|/(k_B T) \ll 1$ (also $v\tau_p \ll \lambda$ for the local expansion). (3) Its population difference is out of phase with the force, so the period average vanishes. (4) Motion creates $\delta f \propto v_z \tau_p$; weighting the force by this lag gives $F_z \propto -v_z$. (5) $V_0 \tau_p$ is intensity independent, but diffusion/friction balance gives $E \sim V_0$. (6) Recoil and localization: the relevant quantum scale is E_r .

4.6 Evaporative Cooling — Minimum Review

Energy selection, collision supply, degeneracy, and runaway evaporation

Mechanism and assumptions. Remove above-average-energy atoms; elastic collisions rethermalize the survivors and refill the high-energy tail. Lower the escape threshold as cooling proceeds. An rf knife selects a resonant surface in a magnetic trap and transfers atoms into an untrapped state. Collisions redistribute energy; escape removes it. Assume a classical, nearly thermal gas, fixed trap shape and strength, constant elastic cross section σ , and a high threshold $\eta \gg 1$ for the rate estimate. Quantum degeneracy eventually limits the classical treatment.

1. Energy bookkeeping (4.94–4.101). Define $E = N\bar{\epsilon}$, escape energy per atom $\epsilon_{\text{escape}} = (1 + \beta)\bar{\epsilon}$, and $dN < 0$.

$$dE = (1 + \beta)\bar{\epsilon} dN = \bar{\epsilon} dN + N d\bar{\epsilon},$$

$$\boxed{d \ln \bar{\epsilon} = \beta d \ln N}.$$

For $V(\mathbf{r}) = K(\hat{\mathbf{r}})r^\nu$, with d confined directions but three kinetic degrees of freedom, put

$$c = \frac{3}{2} + \frac{d}{\nu}, \quad \bar{\epsilon} = ck_B T, \quad \boxed{\frac{d \ln T}{d \ln N} = \beta}.$$

For constant β , $T/T_i = (N/N_i)^\beta$. Positive β means cooling; $\beta = 0$ means loss without cooling. This efficiency measures cooling per fractional atom loss, not per unit time.

2. Useful versus unwanted loss (4.102–4.105).

$$\dot{N}_{\text{ev}} = -\frac{N}{\tau_{\text{ev}}}, \quad \dot{N}_{\text{loss}} = -\frac{N}{\tau_{\text{loss}}}.$$

The simple model assumes unwanted losses carry average energy and add no heat. The useful fraction of all losses is $\tau_{\text{loss}}/(\tau_{\text{loss}} + \tau_{\text{ev}})$. Thus

$$\boxed{\beta' \equiv \frac{d \ln T}{d \ln N} = \beta \frac{\tau_{\text{loss}}}{\tau_{\text{loss}} + \tau_{\text{ev}}}}.$$

Fast evaporation gives $\beta' \approx \beta$; slow evaporation wastes atoms. Per-atom unwanted loss rates scale as n^0 for fixed background gas, n for two-body loss, and n^2 for three-body loss. Three-body recombination allows a third atom to carry away molecular binding energy and momentum. Actual inelastic loss can heat the gas; this is omitted in the simple model.

3. Threshold and collision supply (4.106–4.112). A threshold is a minimum required energy. Here

$$\epsilon = \frac{p^2}{2m} + V(\mathbf{r}) > \epsilon_{\text{ev}}, \quad \eta = \frac{\epsilon_{\text{ev}}}{k_B T}.$$

It refers to total, not kinetic, energy. At high threshold, $\epsilon_{\text{escape}} \approx \epsilon_{\text{ev}}$, so $\beta \approx \eta/c - 1$.

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}}, \quad \bar{v}_{\text{rel}} = \sqrt{2} \bar{v},$$

$$\tau_{\text{el}}^{-1} = n(0)\sigma\bar{v}_{\text{rel}}, \quad \tau_{\text{ev}}^{-1} \simeq \frac{\eta e^{-\eta}}{\sqrt{2} \tau_{\text{el}}}.$$

Here $n(0)$ is the central density; τ_{el} is the book's central elastic-collision time, not a cloud-averaged time. Therefore

$$\boxed{\beta' \simeq \frac{\eta/c - 1}{1 + (\tau_{\text{el}}/\tau_{\text{loss}})(\sqrt{2}/\eta)e^\eta}}.$$

Raising the threshold increases energy removed per escaped atom, but makes escape exponentially rare. To logarithmic accuracy,

$$\eta_{\text{opt}} \sim \ln(\tau_{\text{loss}}/\tau_{\text{el}}).$$

This is a rough optimum, not an exact equality of loss and evaporation times.

4. Phase-space density (4.113). In a fixed trap, $V(R) \sim k_B T$ implies

$$R \propto T^{1/\nu}, \quad \mathcal{V}_r \propto T^{d/\nu}, \quad p_{\text{th}} \propto T^{1/2}, \quad \mathcal{V}_p \propto T^{3/2}.$$

Unconfined spatial dimensions have fixed available size. Thus the occupied phase-space volume scales as T^c and the characteristic occupation per quantum state is

$$\mathcal{D} \propto \frac{N}{T^c}, \quad \boxed{-\frac{d \ln \mathcal{D}}{d \ln N} = c\beta' - 1}.$$

Since $d \ln N < 0$, degeneracy increases when

$$\boxed{\beta' > 1/c}.$$

Cooling must shrink phase space faster, fractionally, than atoms are lost. The momentum exponent is $3/2$, not $d/2$: kinetic motion remains 3D.

5. Runaway evaporation (4.114). With $n \propto N/T^{d/\nu}$ and fixed σ ,

$$\tau_{\text{el}}^{-1} \propto n\sqrt{T}, \quad \boxed{\frac{d \ln \tau_{\text{el}}}{d \ln N} = \beta' \left(\frac{d}{\nu} - \frac{1}{2} \right) - 1}.$$

Runaway means collisions speed up as N falls, so the boxed derivative must be positive. For $d/\nu > 1/2$, this requires $\beta' > 1/(d/\nu - 1/2)$, a stronger condition than degeneracy growth. For a 3D harmonic trap ($d = 3, \nu = 2$):

$$\begin{aligned} T \text{ decreases : } & \beta' > 0, \\ \mathcal{D} \text{ increases : } & \beta' > 1/3, \\ \text{runaway : } & \beta' > 1. \end{aligned}$$

At fixed β' and d , smaller ν improves the phase-space-density gain; this comparison alone does not guarantee an experimentally superior trap.

Evaporative Cooling — Clarifications and Self-check

The steps that were unclear during reading

A. Why $\langle x_i \partial_i V \rangle = k_B T$? The classical position weight is $e^{-V/(k_B T)}$. Integration by parts, with vanishing boundary term, gives in one dimension

$$\begin{aligned}\langle x V'(x) \rangle &= -\frac{k_B T}{Z_x} \int x \frac{d}{dx} e^{-V/(k_B T)} dx \\ &= \frac{k_B T}{Z_x} \int e^{-V/(k_B T)} dx = k_B T.\end{aligned}$$

In several dimensions, integrate over x_i while holding the others fixed; no separability of V is required. Homogeneity gives

$$\sum_{i=1}^d x_i \partial_i V = \nu V \quad \Rightarrow \quad \langle V \rangle = \frac{d}{\nu} k_B T.$$

Add $\langle \epsilon_{\text{kin}} \rangle = 3k_B T/2$ to obtain $\bar{\epsilon} = ck_B T$. Only a quadratic coordinate contributes $k_B T/2$ of potential energy.

B. Why count atoms already above threshold? Use a hypothetical equilibrium gas with no escape. Detailed balance gives $R_{L \rightarrow H} = R_{H \rightarrow L}$, where L and H denote states below and above threshold. Estimate the easier reverse process: energetic atoms colliding and losing their excess energy. Equating their total collision rate to the rate of leaving H is a leading high-threshold approximation, not exact: not every collision moves an atom below threshold. With escape enabled, the nearly thermal low-energy population supplies atoms to H at approximately this reference rate. The evaporating gas itself is not in exact equilibrium.

C. What does the collision integral count? Write $d\Pi_p = d^3p/(2\pi\hbar)^3$. Then

$$\begin{aligned}R_{\text{coll}} &= \int d^3r \int_{\epsilon_p > \epsilon_{\text{ev}}} d\Pi_p f_p \gamma_p, \\ \gamma_p &= \int d\Pi_{p'} f_{p'} \sigma |\mathbf{v}_p - \mathbf{v}_{p'}|.\end{aligned}$$

Here γ_p is the collision frequency of one energetic atom. The two momenta label two incoming atoms, not before/after momenta.

$f_p = \exp[-(p^2/(2m) + V - \mu)/(k_B T)]$ is occupation per state; μ sets normalization. In the dominant collision region, $v_p \gg v_{p'}$, so $v_{\text{rel}} \approx v_p$ and $\int d\Pi_{p'} f_{p'} = n(\mathbf{r})$. Hence

$$R_{\text{coll}} \simeq \sigma \int d^3r n(\mathbf{r}) \int_{\epsilon_p > \epsilon_{\text{ev}}} d\Pi_p f_p v_p.$$

High total energy does not imply high speed everywhere; far up the trap, energetic atoms can be slow, but the partner density is very small.

D. Where do $\eta e^{-\eta}$ and $n(0)$ come from? Put $u = V/(k_B T)$ and $a_v = m/(2k_B T)$. For $u < \eta$, $v_{\text{min}}^2 = 2(\epsilon_{\text{ev}} - V)/m$. The speed-weighted tail is

$$\begin{aligned}\int_H d\Pi_p f_p v_p &= n(\mathbf{r}) \frac{\int_{v_{\text{min}}}^{\infty} v^3 e^{-a_v v^2} dv}{\int_0^{\infty} v^2 e^{-a_v v^2} dv} \\ &= n(\mathbf{r}) \bar{v} (1 + \eta - u) e^{-\eta + u}.\end{aligned}$$

The v^3 combines the spherical measure $v^2 dv$ and collision speed v . The key integral is

$$\int_b^{\infty} x e^{-x} dx = (1 + b) e^{-b}.$$

Since $n = n(0) e^{-u}$, $n^2 e^u = n(0) n$, so the dominant region gives

$$R_{\text{coll}} \simeq n(0) \sigma \bar{v} e^{-\eta} \int d^3r n(\mathbf{r}) (1 + \eta - u).$$

Keeping only the leading term in $\eta \gg 1$ and using $\int n d^3r = N$,

$$R_{\text{coll}} \simeq N n(0) \sigma \bar{v} \eta e^{-\eta}.$$

The region $u > \eta$ is exponentially subleading. Central density emerges from the Boltzmann factors; it was not substituted for density everywhere. Finally $N_{\text{ev}} \simeq -R_{\text{coll}}$ and $n(0) \sigma \bar{v} = 1/(\sqrt{2} \tau_{\text{el}})$ give Eq. (4.111).

E. Why $\bar{v}_{\text{rel}} = \sqrt{2} \bar{v}$? Independent equal-mass thermal velocities each have component variance $k_B T/m$. Their difference has twice that variance, so its Maxwell speed scale, and hence mean speed, is $\sqrt{2}$ larger. The mean speed is nonzero even though the mean velocity vector is zero.

F. Where does the minus sign in (4.113) come from?

$$\ln \mathcal{D} = \ln N - c \ln T + \text{const} \quad \Rightarrow \quad \frac{d \ln \mathcal{D}}{d \ln N} = 1 - c \beta'.$$

Multiplying by -1 defines a positive gain per fractional atom loss. Because $d \ln N < 0$, a negative derivative means increasing $\mathcal{D}$.

Self-check — cover the key.

1. Derive $d \ln T = \beta d \ln N$ from energy conservation.
2. Why is potential energy $dk_B T/\nu$, not always $dk_B T/2$?
3. Why does a higher threshold not always improve actual efficiency?
4. What equilibrium process estimates the supply of escaping atoms?
5. Reconstruct $N n(0) \sigma \bar{v} \eta e^{-\eta}$ from the tail integral.
6. Why are the spatial and momentum exponents d/ν and $3/2$?
7. State the distinct conditions for cooling, degeneracy growth, and runaway in a 3D harmonic trap.

Key. (1) Differentiate $E = N \bar{\epsilon}$ and use escape energy $(1 + \beta) \bar{\epsilon}$, with fixed $\bar{\epsilon}/T$. (2) Integration by parts plus Euler's homogeneity identity. (3) Energy selectivity grows linearly but waiting time grows exponentially; unwanted losses continue. (4) Reverse high-to-low transitions, approximated by energetic-atom collisions. (5) Use the speed-weighted exponential tail, $n^2 e^u = n(0) n$, and $\int n = N$. (6) $K R'' \sim k_B T$ and three thermal momentum widths $\propto \sqrt{T}$. (7) $\beta' > 0$, $\beta' > 1/3$, $\beta' > 1$.

4.7 Spin-Polarized Hydrogen — Minimum Review

Historical experiment: preparation, trapping, cooling, loss, and detection

Main lesson. BEC requires a long-lived atomic gas, sufficient elastic collisions for cooling, and a detectable condensate. Absence of molecular bound states is not required: a metastable gas suffices if loss is slow compared with preparation and observation. The numbers and experimental limitations below describe the historical experiment in the textbook, not universal limits on present techniques.

1. Why hydrogen? Spin-aligned hydrogen has no two-body molecular bound state in the electronic triplet channel. This does not mean ordinary H_2 cannot bind: the strongly bound molecule has a different electronic spin configuration. Alkalis show that molecular bound states need not prevent atomic BEC; slow molecule formation at low density can leave a useful atomic lifetime.

2. Why cryogenic rather than laser cooling? The strong $1S \rightarrow 2P$ Lyman- α line is near 122 nm, making suitable light difficult to produce. Its linewidth sets an unfavorable Doppler scale, while short wavelength and small atomic mass give large recoil:

$$E_r = \frac{\hbar^2 q^2}{2m}, \quad q = \frac{2\pi}{\lambda}.$$

Historical solution: thermal contact with cold walls via helium, followed by thermal isolation and evaporative cooling.

3. What are a, b, c, d ? They are four hyperfine-Zeeman spin substates within the electronic $1S$ ground state, not four different orbitals. Hydrogen has electron spin $s = 1/2$ and proton spin $i = 1/2$. In the strong-field picture, write $|\text{electron, proton}\rangle$, with arrows relative to the field:

state	strong-field spin state	seeking
a	$\simeq \downarrow\uparrow\rangle$	high field
b	$ \downarrow\downarrow\rangle$	high field
c	$\simeq \uparrow\downarrow\rangle$	low field
d	$ \uparrow\uparrow\rangle$	low field

The seeking labels refer to the trapping regime discussed. At finite field, a, c are mixtures of the opposite-spin basis states; b, d are unmixed parallel-spin states. Always determine the force from the actual energy slope:

$$U(\mathbf{r}) = E_{\text{int}}[B(\mathbf{r})], \quad \mathbf{F} = -\frac{dE_{\text{int}}}{dB} \nabla B.$$

Positive slope means low-field seeking. Static magnetic minima trap these states; a field-magnitude maximum is unavailable in the current-free region. Do not equate electron-spin direction with magnetic-moment direction.

Confusion: “doubly polarized.” Both electron and proton spin populations are polarized. A pure d sample has electrons up *and* protons up. A strong-field mixture of c, d can have aligned electron spins but mixed proton spins. Pure b is also doubly polarized, but is high-field seeking. “Double” counts two spin species, not two atoms or two units of spin.

4. Experimental sequence.

1. An rf discharge dissociates molecular hydrogen in the source cell.
2. Low-field-seeking c, d atoms enter an Ioffe-Pritchard magnetic trap.
3. Helium transfers heat from hydrogen to the cold chamber walls.
4. Spin exchange $c + c \rightarrow d + b$ removes c : the b product escapes, while d remains. The sample becomes dominated by d atoms.
5. Further wall cooling makes helium stick to the walls, removing the buffer gas and thermally isolating hydrogen from the environment.
6. Evaporation removes energetic hydrogen; hydrogen-hydrogen collisions rethermalize the survivors.

The source rf discharge and the rf evaporation knife have different roles. Thermal isolation from walls does not mean eliminating interatomic collisions.

5. Weak scattering, high density, and loss.

Hydrogen’s small scattering length implies a small elastic cross section. The rate $\Gamma_{\text{el}} \sim n\sigma\bar{v}_{\text{rel}}$ must remain high enough for evaporation; cryogenic loading supplies relatively high density. Weak interactions simplify theory but can hinder preparation. The phase-space-density criterion uses

$$\lambda_T = \sqrt{\frac{2\pi\hbar^2}{mk_B T}}, \quad \mathcal{D} \sim n\lambda_T^3.$$

Small mass and high density favor degeneracy at higher temperature. The reported onset is about $50\ \mu\text{K}$. Dipolar relaxation, schematically $d + d \rightarrow a + a$, destroys trapped atoms. “Doubly polarized” suppresses important loss channels, not every channel. Density-dependent central loss can preferentially remove low-energy atoms, producing anti-evaporative heating; do not assume all released internal energy is retained by the gas. Loss competes with slow elastic thermalization.

Historical scales to recognize, not memorize. The excerpt reports a condensate fraction near 5%, yet about 10^9 condensed atoms; peak density $\sim 5 \times 10^{15}\ \text{cm}^{-3}$; cloud length $\sim 5\ \text{mm}$ and diameter $\sim 15\ \mu\text{m}$. A small fraction N_0/N does not imply a small population N_0 .

6. Detection by spectroscopy. Use the narrow $1S \rightarrow 2S$ transition with two photons near 243 nm:

$$2h\nu_{\text{laser}} = E_{2S} - E_{1S}.$$

The transition is forbidden as a one-photon electric-dipole process, not forbidden by every interaction. Interactions shift the transition frequency in a density-dependent way, $\Delta\nu \propto n$ in the treatment described. The dense condensate and dilute thermal cloud contribute distinguishable spectral components. This detects density rather than relying on the small condensate fraction in a time-of-flight image.

Hydrogen — Selection Rules and Self-check

Resolving the angular-momentum tensor-product question

A. Orbital notation versus spin. In $n\ell$ notation, the orbital letters S, P, D, F mean $\ell = 0, 1, 2, 3$. Thus $1S$ means $n = 1, \ell = 0$; the letter S here is not electron spin. In the formulas below, L denotes orbital angular momentum, not spin.

B. Energy matching is not enough. In the electric-dipole (E1) approximation the interaction is proportional to $\mathbf{E} \cdot \mathbf{r}$. Its transition amplitude contains

$$M_q = \langle n' L' m' | r_q | n L m \rangle,$$

where r_q is a spherical component of the position operator. An energy-matched photon cannot drive an E1 transition if this matrix element vanishes. “Allowed” by a selection rule means not excluded by that rule; it does not guarantee a nonzero radial overlap or a strong line.

C. Rotations alone permit the middle channel. The position operator is a rank-one spherical tensor. For $L \geq 1$,

$$D^{(1)} \otimes D^{(L)} = D^{(L-1)} \oplus D^{(L)} \oplus D^{(L+1)}.$$

This is a *tensor product decomposed into a direct sum*, not a direct sum of the original vector spaces. Wigner–Eckart gives

$$M_q \propto C_{Lm, 1q}^{L'm'} \langle n' L' || r || n L \rangle.$$

Rotational symmetry permits $L' = L - 1, L, L + 1$ and $m' = m + q$. Other symmetries may force the reduced matrix element to vanish.

D. Parity excludes $L' = L$ for one-electron orbitals.

Let $\mathcal{P}$ denote inversion. Then

$$\mathcal{P} |n L m\rangle = (-1)^L |n L m\rangle, \quad \mathcal{P} r_q \mathcal{P}^{-1} = -r_q.$$

Inserting parity on both sides of the matrix element gives

$$M_q = (-1)^{L'+L+1} M_q.$$

For a nonzero matrix element, $L' + L$ must be odd.

Combining this with the rotation rule leaves

$$\Delta L = \pm 1 \quad \text{for hydrogen orbital E1 transitions.}$$

For example, two S orbitals are even under inversion while $\mathbf{r}$ is odd, so opposite spatial points cancel in $\int \psi_{2S}^* \mathbf{r} \psi_{1S} d^3r$.

E. The $SO(3)$ versus $O(3)$ explanation. If a

representation is labeled by angular momentum and parity (L, π) , then the dipole has labels $(1, -1)$. For $L \geq 1$,

$$(1, -1) \otimes (L, \pi) = (L - 1, -\pi) \oplus (L, -\pi) \oplus (L + 1, -\pi).$$

The middle channel survives as a representation, but it has *opposite* parity. A one-electron orbital with the same L has the original parity $(-1)^L$, so it cannot realize that channel. The user’s tensor-product argument was correct; the missing constraint was parity.

F. Two important qualifications.

1. For $L = 0$, the correct product is

$$D^{(1)} \otimes D^{(0)} = D^{(1)}.$$

There is no scalar channel: $S \rightarrow S$ is already excluded by rotations for one E1 photon, independently of the parity check.

2. For many-electron atoms, total parity is not fixed by total orbital L : configurations with the same L can have opposite parity. Thus $\Delta L = 0$ can be E1-allowed in LS coupling, except $L = 0 \rightarrow 0$. For total angular momentum J , the general rank-one rule similarly allows $\Delta J = 0, \pm 1$, excluding $J = 0 \rightarrow 0$, with a parity change. Do not apply the one-electron $\Delta \ell = \pm 1$ rule blindly to total L or J .

G. Why two photons permit $1S \rightarrow 2S$. Two E1 interactions can connect $S \rightarrow P \rightarrow S$. Schematically, the second-order amplitude contains a sum over intermediate P states j :

$$\mathcal{A}_{fi}^{(2)} \propto \sum_j \frac{\langle f | \mathbf{e}_2 \cdot \mathbf{r} | j \rangle \langle j | \mathbf{e}_1 \cdot \mathbf{r} | i \rangle}{E_i + \hbar\omega_1 - E_j} + (1 \leftrightarrow 2).$$

Here $\mathbf{e}_1, \mathbf{e}_2$ are polarization vectors and ω_1, ω_2 are photon angular frequencies. Energy conservation requires $\hbar(\omega_1 + \omega_2) = E_f - E_i$. The intermediate contribution is virtual, not a resonantly occupied P level after the first half-energy photon. Two odd-parity operators have even combined parity, and $1 \otimes 1$ contains a scalar channel: an $S \rightarrow S$ transition is therefore possible. Individual photon polarizations still matter.

Self-check — cover the key.

1. Distinguish a, b, c, d from $1S, 2S, 2P$.
2. What is doubly polarized? Why are a, c arrow states approximate?
3. Why remove helium before evaporation, but retain hydrogen collisions?
4. Why does small scattering length make high density useful?
5. How does a spectral shift distinguish condensate from thermal atoms?
6. Why does $1 \otimes L$ not by itself imply $\Delta L = \pm 1$?
7. What happens to the middle L channel after parity is included?
8. Explain both the $L = 0$ exception and the many-electron qualification.
9. Why is $S \rightarrow P \rightarrow S$ not a sequential resonant excitation here?

Key. (1) Spin substates within the ground orbital versus electronic orbital labels. (2) Electron and proton polarization; hyperfine coupling mixes opposite-spin configurations. (3) Break thermal contact with walls; retain rethermalization. (4) Compensate small σ in $n\sigma\bar{v}$. (5) Different densities give different interaction shifts. (6) Rotations also allow $L' = L$; a reduced matrix element may vanish. (7) It requires opposite parity, absent for same- L hydrogen orbitals. (8) $1 \otimes 0 = 1$; many-electron parity need not equal $(-1)^L$. (9) Each photon is off resonance with the intermediate P states; only their summed energy matches the final excitation.